Lecture 4: Mapping between Spaces

Seonyeong Park, Ph.D. (: sp33@illinois.edu)
Research Assistant Professor
Bioengineering
University of Illinois Urbana-Champaign
February 3, 2025

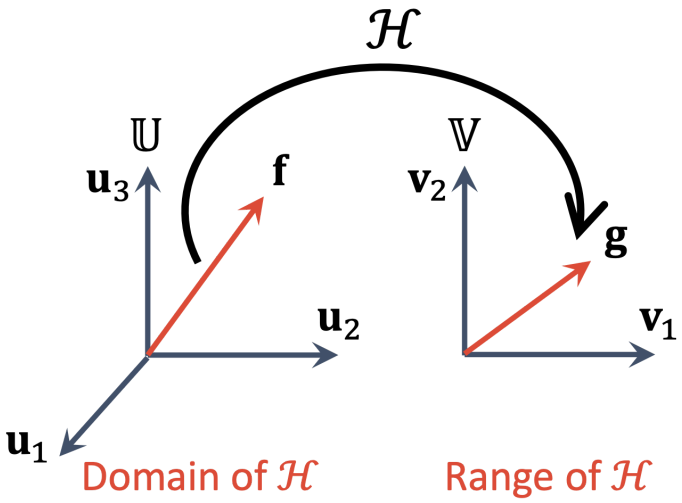
🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Continuous-to-continuous, continuous-to-discrete, and discrete-to-discrete mappings
- Injective, surjective, and bijective mappings
- Null space

Operators

Operators, denoted by $\mathcal{H}$, map from one space to another and are expressed as $\mathbf{g} = \mathcal{H}\mathbf{f}$



- ***Continuous-to-continuous (C-C)*** mapping: Maps a continuous vector (i.e., function) in one space to a continuous vector in another space
  - **Input**: $f(x) \in \mathbb{L}_2$, **Output**: $g(x') \in \mathbb{L}_2$

- ***Continuous-to-discrete (C-D)*** mapping: Maps a continuous vector (i.e., function) in one space to a discrete vector in another space
  - **Input**: $f(x) \in \mathbb{L}_2$, **Output**: $\mathbf{g} \in \mathbb{R}^M$

- ***Discrete-to-discrete (D-D)*** mapping: Maps a discrete vector in one space to a discrete vector in another space
  - **Input**: $\mathbf{f} \in \mathbb{R}^N$, **Output**: $\mathbf{g} \in \mathbb{R}^M$

❓ What are the examples of C-C, C-D, and D-D mappings?

- **C-C mapping**: Fourier transform
- **C-D mapping**: Digital imaging with real-world objects
- **D-D mapping**: Digital imaging with digital/virtual/synthetic objects, image reconstruction, and image processing

Mathematical Descriptions for $\mathcal{H}$

$$g(x') = \int_{\alpha}^{\beta} dx\, h(x', x)\, f(x) \quad \textbf{C-C mapping, integral transforms}$$

$$g_m = \int_{\alpha}^{\beta} dx\, h_m(x)\, f(x) \quad \textbf{C-D mapping}$$

$$g_m = \sum_{n=1}^{N} H_{mn} f_n \quad \textbf{D-D mapping, matrix operators}$$

$h(x', x)$ and $h_m(x)$ are kernels that define the mapping relationship.

Matrix Operator

Suppose $\mathbf{f} \in \mathbb{R}^N$, $\mathbf{g} \in \mathbb{R}^M$, and let the matrix operator $\mathbf{H} \in \mathbb{R}^{M \times N}$ represent the mapping $\mathbf{H} : \mathbb{R}^N \to \mathbb{R}^M$. The mapping rule follows the standard matrix multiplication:

$$\begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_m \\ \vdots \\ g_M \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1n} & \dots & h_{1N} \\ h_{21} & h_{22} & \dots & h_{2n} & \dots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{m1} & h_{m2} & \dots & h_{mn} & \dots & h_{mN} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \dots & h_{Mn} & \dots & h_{MN} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \\ \vdots \\ f_N \end{bmatrix}$$

For the m-th element of the mapping:

$$g_m = \sum_{n=1}^{N} H_{mn} f_n$$

In matrix form:

$$\mathbf{g} = \mathbf{H}\mathbf{f}$$

Linearity of the Mapping

The mapping defined by $\mathbf{H}$ is linear, meaning it satisfies the following property:

$$\mathbf{H}(\mathbf{f}_1 + \alpha \mathbf{f}_2) = \mathbf{H}\mathbf{f}_1 + \alpha \mathbf{H}\mathbf{f}_2, \quad \alpha \in \mathbb{R}.$$

Rank of $\mathbf{H}$

The rank of $\mathbf{H}$ is the dimension of the vector space spanned by its rows (row rank) or columns (column rank).

It represents the number of linearly independent measurements.

- Row rank: The number of linearly independent rows of $\mathbf{H}$
- Column rank: The number of linearly independent columns of $\mathbf{H}$

The rank follows the rule:

$$\text{rank}(\mathbf{H}) \leq \min(M, N)$$

In [8]:
```python
import numpy as np

# Define a D-D operator H
H = np.array([[1, 1, 0],
              [1, 1, 0],
              [2, 3, 1]])

# Compute the rank of the matrix H
rank_H = np.linalg.matrix_rank(H)
```

```
# Display the matrix and its rank
print(f"Operator (Matrix H):\n {H}")
print(f"Rank of H: {rank_H}")
```

```
Operator (Matrix H):
 [[1 1 0]
 [1 1 0]
 [2 3 1]]
Rank of H: 2
```

Hilbert Space Operators

Let $\mathbb{U}$ and $\mathbb{V}$ be Hilbert spaces, where:

- $\mathbf{f} \in \mathbb{U}$: The object to be imaged
- $\mathbf{g} \in \mathbb{V}$: The resulting (measured) data
- $\mathcal{H}$: A linear operator such that:

$$\mathbf{g} = \mathcal{H}\mathbf{f}$$

Range and Domain

- Domain of $\mathcal{H}$: The space $\mathbb{U}$, where all possible $\mathbf{f}$ reside
- Range of $\mathcal{H}$: A subspace of $\mathbb{V}$, where all possible $\mathbf{g}$ lie through the mapping by $\mathcal{H}$

Inner Product in $\mathbb{U}$ and $\mathbb{V}$

- Inner product in $\mathbb{U}$: $\langle \mathbf{f}_1, \mathbf{f}_2 \rangle_{\mathbb{U}}$
- Inner product in $\mathbb{V}$: $\langle \mathbf{g}_1, \mathbf{g}_2 \rangle_{\mathbb{V}}$

Norm in $\mathbb{U}$ and $\mathbb{V}$

- Norm in $\mathbb{U}$: $\|\mathbf{f}\| = \sqrt{\langle \mathbf{f}, \mathbf{f} \rangle_{\mathbb{U}}}$
- Norm in $\mathbb{V}$: $\|\mathbf{g}\| = \sqrt{\langle \mathbf{g}, \mathbf{g} \rangle_{\mathbb{V}}}$

Linearity

The operator $\mathcal{H}$ is linear if, for all vectors in its domain and any scalar c, it satisfies:

$$\mathcal{H}[\mathbf{f}_1 + c\mathbf{f}_2] = \mathcal{H}\mathbf{f}_1 + c\mathcal{H}\mathbf{f}_2.$$

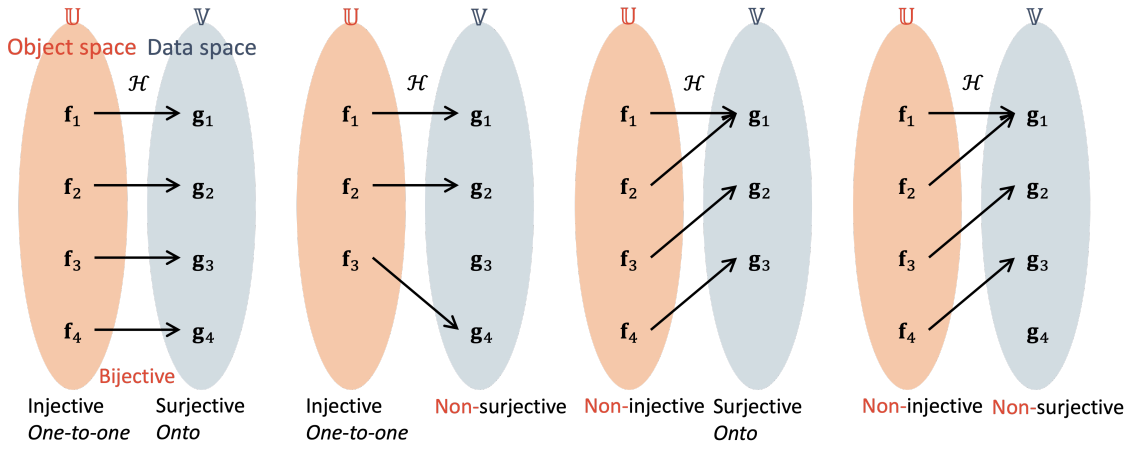
Mapping between Spaces

Let $\mathbb{U}$ be the object space and $\mathbb{V}$ be the data/image space, both of which are Hilbert spaces.

Suppose $\mathbf{f}_n \in \mathbb{U}$ is an object vector and $\mathbf{g}_m \in \mathbb{V}$ is a (measured) data vector.

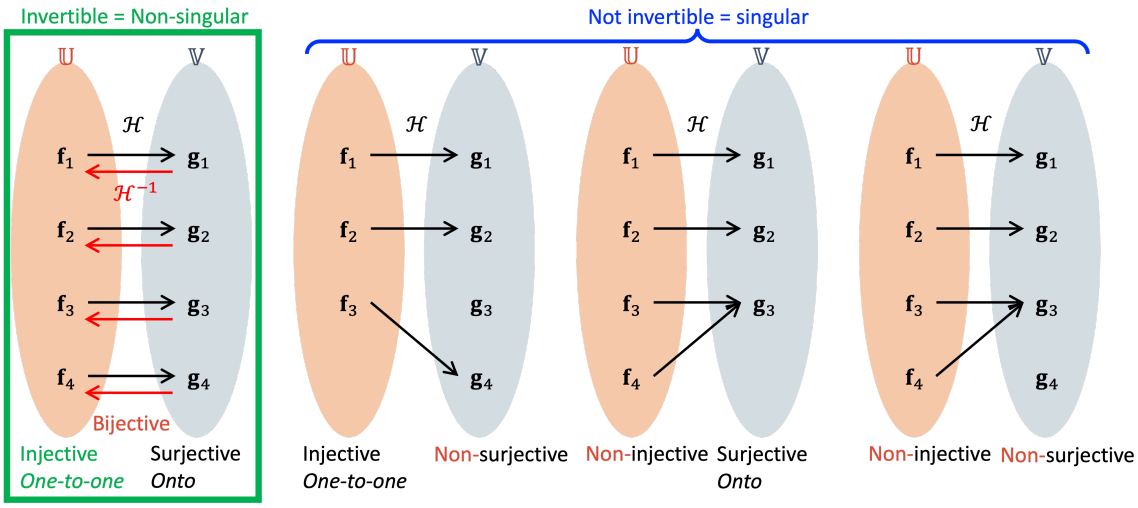
Mapping Types

- One-to-one (injective): Each $\mathbf{f}_n$ maps to a unique $\mathbf{g}_m$
- Not one-to-one (non-injective): Multiple $\mathbf{f}_n$ may map to the same $\mathbf{g}_m$
- Onto (surjective): Every $\mathbf{g}_m$ is mapped from at least one $\mathbf{f}_n$
- Non onto (non-surjective): Some $\mathbf{g}_m$ are not mapped from any $\mathbf{f}_n$



- Bijective: Each $\mathbf{f}_n$ maps uniquely to every $\mathbf{g}_m$
  - One-to-one (injective) and onto (surjective)
  - Invertible: An inverse operator exists.
  - Nonsingular: The operator has no null space.

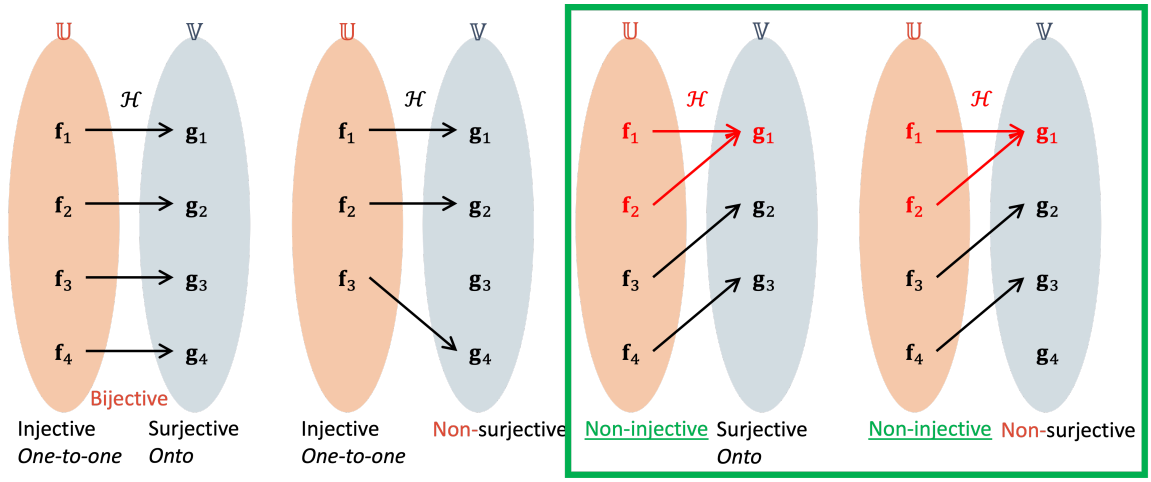
⭐ *Note: Real-world imaging operators (unlike computer simulations) are generally singular (non-bijective).*



Null Vector and Null Space of $\mathcal{H}$

Assume a **non-injective (not one-to-one)** operator $\mathcal{H}$, where two different vectors $\mathbf{f}_1$ and $\mathbf{f}_2$ map to the same vector $\mathbf{g}_1$:

$$\mathbf{g}_1 = \mathcal{H}\mathbf{f}_1 \text{ and } \mathbf{g}_1 = \mathcal{H}\mathbf{f}_2$$



$$\mathcal{H}\mathbf{f}_2 - \mathcal{H}\mathbf{f}_1 = \mathbf{g}_1 - \mathbf{g}_1$$
$$\mathcal{H}(\mathbf{f}_2 - \mathbf{f}_1) = \mathbf{0}$$
$$\mathcal{H}\mathbf{f}_{null} = \mathbf{0}$$

where $\mathbf{f}_{null}$ is a null vector of $\mathcal{H}$, meaning it belongs to the null space (a subspace of $\mathbb{U}$) of the operator $\mathcal{H}$.

⭐ *Note: Only one-to-one (injective) operators have no null space.*